

Sanjan Baitalik

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Research Interests

Trustworthy ML (xAI + robustness), Computer Vision (Hyperspectral/Multimodal), Human-Computer Interaction

Education

Institute of Engineering & Management(IEM), Kolkata

August 2022 – Expected July 2026

B.Tech in Computer Science & Engineering

([Transcripts](#)) CGPA: 9.33/10

Awarded the **Chancellor's Award for Exemplary Research Contribution**, one of only two students selected in the department. Ranked among the **top 10%** of the class by CGPA.

Experience

University of Nebraska-Lincoln, USA

June 2025 – Present

Research Intern

Remote

Supervisor: *Dr. Sruti Das Choudhury* ([Offer Letter](#))

- Co-authored a human-centered XAI study that combined k-means/hierarchical clustering, SHAP-driven interpretability, and narrative visualization to turn a 22-crop agronomic dataset and a 500-record/6-variable hospital-cost cohort into 4 actionable clusters (e.g., crop regimes sized 797/618/585/200), demonstrating how rescaling/categorical handling improves cluster trustworthiness for decision-making [P3].

- Co-authored *ReproPheno/ReproPhenoNet*, introducing a large-scale UNL reproductive-stage plant phenotyping benchmark (~6 TB) with multiview, temporal visible-light, fluorescence, and hyperspectral image sequences of flowers/fruits, and developed a YOLO-based ReproPhenoNet pipeline for flower/fruit detection and temporal phenotype extraction, achieving combined mAP@0.5=0.849 and F1=0.838 [P7].

- Co-developed *PlantPhenoLM*, a retrieval-augmented LLM reasoning framework for phenotype-to-genotype inference that fuses classifier probabilities, nearest-neighbor phenotype evidence, entropy-based ambiguity diagnostics, and selective prediction to produce auditable genotype reports; achieved top-5 recovery of 0.952 and improved retrieval-augmented top-1 accuracy to 0.429 on 5-fold held-out evaluation [P8].

- Built an interactive visual-analytics pipeline on UNL greenhouse phenotyping data (42 plants, 9 genotypes, 25 days) that couples temporal embeddings, DTW-based clustering, and SHAP/LIME-linked causal views, achieving ARI=0.30 and NMI=0.62 genotype-cluster agreement while enabling “early prediction” analysis via accuracy-vs-day curves [P14].

- Engineered HyperProbe, a lightweight Streamlit-based human-in-the-loop hyperspectral tool spanning 517-1700 nm (243 bands) that integrates pixel/ROI selection, derivative + spectral-angle analytics, Otsu band-difference segmentation, interactive MLP/LR/RF classification, clustering, and PCA/t-SNE embeddings into a single end-to-end workflow [P15].

- Featured in the university's news story snr.unl.edu (August, 2025).

University of Calcutta

January 2025 – December 2025

Research Scholar

Kolkata, India

Supervisors: *Dr. Arup Kumar Chattopadhyay, Prof. Amit Kumar Das, Prof. Amlan Chakrabarti*

- Developed and benchmarked FH-FAM, a fuzzy-hypergraph feature-selection algorithm for high-dimensional agriculture & remote-sensing data, achieving 81.43% mean classification accuracy with 89.28% mean feature reduction across 15 public datasets, outperforming multiple established baselines [P1].

- We propose SIF-HFAM, a strong intuitionistic fuzzy hypergraph framework equipped with a monotone submodular coverage objective. The method admits the classical greedy $(1 - 1/e)$ -approximation guarantee and achieves an average accuracy of approximately 78% while removing ~98% of the features (retaining < 2%) across 14 high-dimensional benchmark datasets, all with favorable computational runtime [P4].

Generative AI CoE, IEM

November 2024 – Present

Student Research Lead at GenAI CoE

Kolkata, India

Established and led GenAI CoE's research sub-committee, driving its end-to-end execution and operations across projects, member onboarding, 10+ journal teams, website management, and initiatives including *ReelBook* (Pearson collaboration) and *Medium publishing* to scale research output and AI training across IEM.

IEM Research Foundation

August 2024 – March 2025

Project Intern at bair.ai ([Certificate](#))

Kolkata, India

Built *MemeMetric*, an end-to-end cluster-based cryptocurrency forecasting system by architecting the full data/ML pipeline and automated reporting, and integrated real-time Twitter/Telegram/Reddit sentiment signals via NLP to strengthen robustness and reduce forecast error/volatility.

Innovation & Entrepreneurship Development Cell (CSE)

March 2024 – August 2024

Undergraduate Research Assistant ([Certificate](#))

Kolkata, India

Co-authored an IEM-HEALS 2024 accepted paper by modeling Jul 2019–Dec 2022 closing prices of 20 pharma stocks via multivariate regression, volatility, and event-study analysis, and built *TraderBot*, a Flask+MongoDB real-time trading simulator integrated with Yahoo Finance for live strategy and portfolio testing [P13].

National University of Singapore (NUS)

July 2023 (1 week)

Undergraduate Study Abroad Program ([Certificate](#))

Singapore

Studied fundamentals of “Artificial Intelligence, Internet of Things, Machine Learning & Data Analytics”, lectured by [Dr. Peter Leong](#), [Dr. Eric Cambria](#), [Dr. Matthew Chua](#), [Dr. Yiliang Zhao](#), [Dr. Gábor Benedek](#), [Dr. Tan Kian Hua](#), [Yong Heng Michael Tan](#), [Marton Szel](#), [Gillian Cheng](#).

Selected Independent Research

- Co-developed *MiQ-MCP*, a conditionally robust uncertainty-quantification framework for high-frequency financial time series that combines Fourier-enriched quantile regression with Mondrian conformal calibration over intraday time buckets, producing valid prediction intervals for minute-scale NIFTY-50 forecasting under strong time-of-day volatility shifts [P2].
- Investigated garden-path recovery in causal and masked language models using 100 English garden-path/control pairs across NP/Z, NP/S, and MV/RR constructions, showing that decoder-only models exhibit sharper online syntactic revision signals while masked encoders appear more buffered due to bidirectional context access [P5].
- Co-built *PolyJudge-Uncertain*, a 5,120-item multilingual benchmark spanning English, Hindi, Hinglish and Bengali to study hedging effects in LLM-as-a-judge evaluation; found that pointwise hedging penalties largely disappear after template repair, but pairwise judges overwhelmingly prefer assertive answers under semantic equivalence [P6].
- Proposed *GreedySAT Revision*, a neuro-symbolic framework that treats multi-turn LLM dialogue as belief revision, using an external SAT/SMT-style consistency checker to maintain globally coherent belief states and eliminate inconsistent final dialogue states under adversarial stress settings [P9].

Publications

Published/Accepted

Journals

- P1. [Rajashik Datta](#), **Sanjan Baitalik**, [Sruti Das Choudhury](#), [Arup Kumar Chattopadhyay](#), [Amit Kumar Das](#), “Fuzzy Hypergraph Feature Association Map for High-Dimensional Feature Selection in Agriculture and Remote Sensing”, *International Journal of Fuzzy Systems*, 2026.
- P2. **Sanjan Baitalik**, [Rajashik Datta](#), [Darothi Sarkar](#), [Ayan Chaudhuri](#), “MiQ-MCP: Valid and Conditionally Robust Uncertainty Quantification for High-Frequency Financial Time Series via Mondrian Conformalized Quantile Regression”, *Computational Economics*, 2026.
- P3. [Sruti Das Choudhury](#), [Rajashik Datta](#), **Sanjan Baitalik**, “Enhancing Interpretability Through Clustering, Explainable AI, and Narrative Visualization: Applications in Precision Agriculture and Healthcare Patient Segmentation”, *Digital Engineering*, 2026.
- P4. **Sanjan Baitaik**, [Rajashik Datta](#), [Arup Kumar Chattopadhyay](#), [Amit Kumar Das](#), [Amlan Chakraborty](#), “From Graphs to Hypergraphs: Submodular Coverage-Based Feature Selection on Intuitionistic Fuzzy Hypergraphs (SIFHFAM)”, *Pattern Recognition*, 2026 (Under Revision).

Conferences

- P5. **Sanjan Baitalik**, [Rajashik Datta](#), “Garden Path Recovery in Causal and Masked Language Models”, *ACL Student Research Workshop*, 2026.
- P6. [Rajashik Datta](#), **Sanjan Baitalik**, “Confidence as a Tie-Breaker: Reassessing Multilingual Hedging Bias in LLM-as-a-Judge Evaluation”, *ACL Student Research Workshop*, 2026.
- P7. **Sanjan Baitalik**, [Rajashik Datta](#), [Utsho Banerjee](#), [Rajarshi Karmakar](#), [Vincent Stoerger](#), [Himadri Nath Saha](#), [Sruti Das Choudhury](#), “ReproPheno and ReproPhenoNet: A Large-Scale Multimodal Benchmark Dataset and Deep Learning Framework for Reproductive-Stage Plant Phenotyping”, *AAAI AgriAI Workshop*, 2026.
- P8. [Rajashik Datta](#), **Sanjan Baitalik**, [Amit Kumar Das](#), [Sruti Das Choudhury](#), “PlantPhenoLM: Phenotype-Genotype Mapping Inference with Multi-Turn LLM Reasoning and Selective Prediction”, *AAAI Bridge on Logic & AI*, 2026.
- P9. **Sanjan Baitalik**, [Rajashik Datta](#), [Amit Kumar Das](#), [Sruti Das Choudhury](#), “Conversation as Belief Revision: GreedySAT Revision for Global Logical Consistency in Multi-Turn LLM Dialogues”, *AAAI Bridge on Logic & AI*, 2026.
- P10. [Sanket Ghosh](#), **Sanjan Baitalik**, [Rajashik Datta](#), [Romit Mukherjee](#), [Darothi Sarkar](#), [Ayan Chaudhuri](#), “Explanation-First Agentic Forecaster for Stock Market”, *International Conference on Electronics, Materials Engineering and Nano-Technology (IEMENTech 2026)*, 2026.
- P11. **Sanjan Baitalik**, [Rajashik Datta](#), [Sanket Ghosh](#), [Darothi Sarkar](#), [Ayan Chaudhuri](#), “Machine Learning-Driven Insights For Stock Market Analysis And Trading”, *International Conference on Interdisciplinary Research in Technology and Management (IRTM 2024)*, 2024.

P12. [Sanket Ghosh](#), [Sanjan Baitalik](#), [Rajashik Datta](#), [Darothi Sarkar](#), “The COVID-19 Shock: An Analysis Of Impacts And Responses Of Indian Stock Market”, [International Conference on Interdisciplinary Research in Technology and Management \(IRTM 2024\)](#), 2024.

P13. [Rajashik Datta](#), [Sanjan Baitalik](#), [Sanket Ghosh](#), [Saugata Ghosh](#), [Swarnendu Ghosh](#), “Is Indian Financial Market Ready for Pandemics?”, [International Conference on Advancing Science and Technologies in Health Science \(IEM-HEALS 2024\)](#), 2024 [Book of Abstracts](#).

Manuscripts in Preparation

P14. [Rajashik Datta](#), [Sanjan Baitalik](#), [Sruti Das Choudhury](#), [Amit Kumar Das](#), “Visual Analytics of Plant Phenotype–Genotype Dynamics via Temporal Embeddings”. Intended for submission to [IEEE Transactions on Visualization and Computer Graphics](#).

P15. [Sanjan Baitalik](#), [Rajashik Datta](#), [Sruti Das Choudhury](#), “HyperProbe Insight: An Interactive Tool for Exploration of Hyperspectral Image Sequences”. Intended for submission to [IEEE Transactions on Visualization and Computer Graphics](#).

P16. [Rajashik Datta](#), [Sanjan Baitalik](#), [Sruti Das Choudhury](#), “Phenotype-Grounded Temporal Storytelling with Auditable Progression Graphs”. Intended for submission to [Computers and Electronics in Agriculture](#).

Skills & Activities

Programming: Python, Java, C, MATLAB	ML/AI: PyTorch, TensorFlow, Scikit-learn; Transformers
XAI: SHAP, LIME	Data: Pandas, NumPy
Tools: L ^A T _E X, Git, Docker, Jupyter, TensorBoard	Cloud: Google Cloud (Cloud Run/Compute), AWS (S3/EC2)
Activities: GenSpark 1.0 Ideathon (Organizer; coordinated 50+ teams; shortlisted 3 funded ideas), Jun–Aug 2025; ICDC 2025 (Conference volunteer), Apr 2025; Dept. of CSE, IEM (Supported NBA accreditation documentation), Mar 2024	

Projects

Geodesic optimal transport (Transfer Geometry) | [GitHub](#)
Implemented sliced-Wasserstein OT diagnostics on frozen ResNet-18 features and benchmarked across 48 transfer settings (CIFAR-10/STL-10/SVHN), showing strong correlations with zero-shot transfer (Pearson $r \approx -0.71$) and low-data adaptation speed (Spearman $\rho \approx 0.60$ at 200-shot).

Grokking + LoRA (Low-Rank Tax) | [GitHub](#)
Built controlled modular-addition experiments (p=97) comparing full-parameter vs. LoRA-on-frozen-base training for 15k epochs, reproducing classic grokking in the full MLP (99% train @ 288 → 99% val @ 481) and quantifying rank/LR thresholds (e.g., $r=8$ fails to fit; $r=16$ reaches high acc without 99% val within budget).